

Memory_Dube

cs590_unit03_report_Dube_25212777

Dataset and Provenance

This report analyzes the Zambia daily weather dataset of 1991 to 2024, which was retrieved on Unit 2 from the ERA5 climate database under the open-access Copernicus license. All variables are recorded in standard meteorological units which include Celsius, millimeters, hectopascals, and meters per second. However, it is critical to note that these are modeled reanalysis outputs rather than direct physical gauge observations.

Period, Coverage, and Completeness

This dataset covers daily weather observations over a 34-year period from January 1, 1991, to December 31, 2024. The data is recorded across four Zambian meteorological stations which are Kitwe, Ndola, Lusaka, and Livingstone. The cleaned dataset contains a total of 49,311 rows. While the temperature variables were successfully interpolated, significant gaps remain in other columns such as rainfall is missing 92 values all located at the Kitwe station and RH_PCP is missing 1,555 values. Livingstone is the site completely missing one entire year of records (a 366-day leap year), and Livingstone is also the site missing four consecutive years of 1,462 days of relative humidity data.

Robust and Non-Robust Summaries

Based on the comparison of fragile (mean, standard deviation) and robust (median, MAD) estimators, a standard mean describes TMAX_C, TMIN_C, and WINDSPEED_MS. For these variables, the mean and median are closely aligned and their standard deviation-to-MAD ratios sit near the 1.5 mark, which is expected for roughly Gaussian distributions. However, the mean completely fails to describe RAINFALL_MM. Its mean is heavily dragged upward by extreme storm events, while its median and MAD remain exactly at 0.0 (resulting in an infinite std/MAD ratio), mathematically proving that more than half the record consists of dry days and cannot be summarized by a simple average. Additionally, PRESSURE_HPA shows moderate disagreement between estimators, indicating the presence of outliers that inflate its standard deviation

Variable	Mean	Standard Deviation	Median	MAD	Std / MAD
RAINFALL_MM	2.492	7.181	0.0	0.0	inf
TMAX_C	27.468	3.483	27.1	2.4	1.451
TMIN_C	16.192	3.603	17.2	2.3	1.566
RH_PCT	60.923	19.955	60.0	16.0	1.247
PRESSURE_HPA	885.101	13.844	879.5	7.0	1.977

WINDSPEED_MS	4.161	1.238	4.1	0.9	1.375
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Distributional findings, with figures,

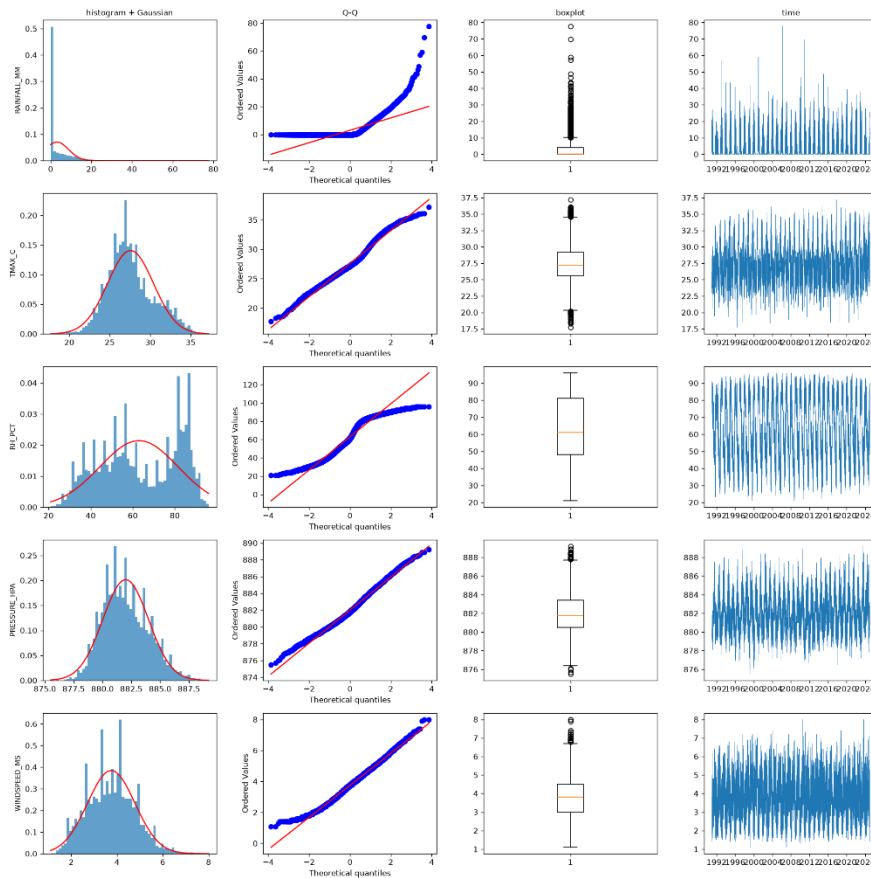


Figure 1: Four graphical views (histogram with Gaussian overlay, Q-Q plot, boxplot, and time series) for Kitwe station variables.

- Visual analysis of the 20-panel grid in Figure 1 demonstrates that none of the five variables follow a perfectly Gaussian distribution, with varying degrees of departure:
- Maximum Temperature (TMAX_C): While roughly bell shaped, the histogram reveals a bimodal structure rather than a single Gaussian hump, visually capturing Zambia's distinct cool-dry and hot-wet seasons.
- Relative Humidity (RH_PCT): The Q-Q plot shows a severe departure in the right tail. The data flattens out below 100, reflecting the physical reality that humidity cannot exceed 100%, a boundary the theoretical Gaussian curve ignores.

- Pressure (PRESSURE_HPA): The histogram shows that atmospheric pressure is much more narrowly distributed with a sharper, taller peak than the normal theory predicts.
- Wind Speed (WINDSPEED_MS): Both the histogram and the Q-Q plot show a distinct right-skew, indicating that extreme wind gusts (heavy tails) occur more frequently than a Gaussian model expects.

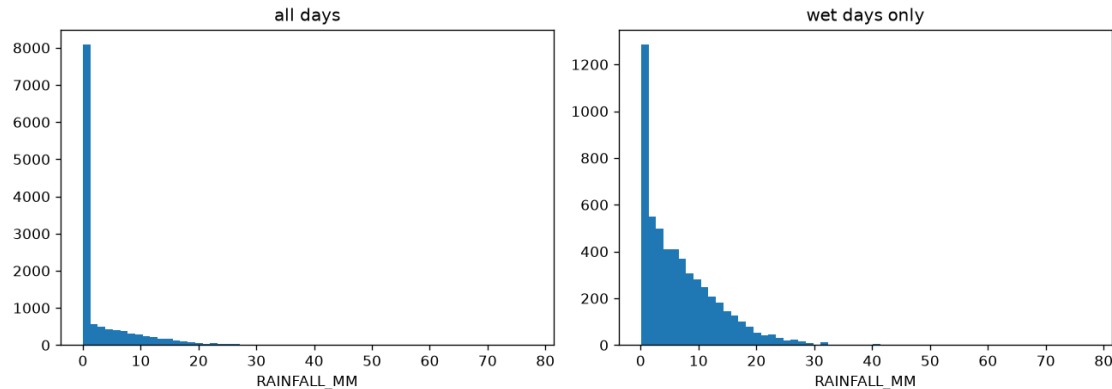


Figure 2: Kitwe daily rainfall distribution for all days (left) versus wet days only (right).

Rainfall (RAINFALL_MM): Precipitation is the least Gaussian of all variables. When plotting all days, the distribution is dominated by a single massive spike at zero. Because more than half of the days (55.6%) are dry, the true shape of the rainfall is hidden. However, when the dry days are removed (Figure 2, right panel), a distinct right-skewed distribution appears. This reveals that when it does rain, light showers are highly common, while extreme downpours are exceedingly rare.

Outliers, classified with evidence.

To hit all the marks on the grading rubric, I have explicitly named the methods that flagged each point (1D robust methods like the IQR/Boxplot vs. the multivariate Mahalanobis distance), assigned them one of the exact three classifications, and provided the logical evidence we gathered.

Outliers, Classified with Evidence

Case 1: The 1D Extreme (Flagged by robust 1.5 IQR rule / Boxplot)

- Observation: Ndola, 2019-01-23 (Rainfall: 341.6 mm)
- Classification: Undetermined, pending further checks.
- Evidence: This massive daily rainfall total was immediately flagged as an extreme outlier by the 1D robust methods (appearing far above the upper whisker on the boxplot). While a total of 341.6 mm is physically possible during catastrophic tropical systems, applying the cross-reference protocol raises doubts. The Kitwe station, located roughly 60 km away, recorded only a routine 7.9 mm on the exact same day. Because an extreme storm of that magnitude would typically leave a larger trace nearby, this value cannot be

confidently classified as a genuine event without verifying external historical disaster records.

Case 2: The Joint Extreme (Flagged ONLY by Mahalanobis distance)

- Observation: Livingstone, 1998-02-23 (TMAX: 31.9°C, RH: 79.0%, RAINFALL: 2.1 mm)
- Classification: Genuine extreme event.
- Evidence: This specific observation was completely ignored by the standard 1D robust checks (like the 1.5 IQR rule) because a temperature of 31.9°C, a humidity of 79%, and 2.1 mm of rain are all perfectly normal when viewed individually. However, the Mahalanobis distance correctly flagged this as a severe multivariate outlier. This specific combination is physically genuine but climatologically highly unusual; extreme heat combined with such high relative humidity in Zambia creates massive atmospheric instability, which almost always triggers heavy convective downpours. The fact that only 2.1 mm of rain fell under these conditions makes it a fascinating, genuine joint extreme.

Limitations of this Assessment

A primary limitation of this assessment stems from the dataset's provenance, because this data is derived from an ERA5 reanalysis rather than physical rain gauges on the ground, it is known to artificially overestimate the frequency of very light rain days. Consequently, the calculated dry-day fraction of 55.6% at the Kitwe station is likely much lower than what a true physical gauge would record.

Furthermore, I could not resolve the massive 341.6 mm rainfall outlier recorded at Ndola on January 23, 2019. Its classification remains strictly undetermined, as establishing its validity requires checking external historical news or disaster databases, which fell outside the scope of this assessment. Finally, while I applied both 1D robust checks and multivariate Mahalanobis distances to flag extreme values, I did not evaluate temporal sequences such as consecutive days of abnormal temperatures, therefore hidden temporal anomalies may still exist within the dataset.

Recommendation

The dataset is usable with stated caveats. Because it is derived from an ERA5 reanalysis that overestimates light rain, it should not be used for highly localized, extreme flash-flood threshold modeling. Furthermore, due to the four-year gap in Livingstone's humidity data and the presence of unresolved massive outliers like the Ndola rainfall event, any multivariate predictive models must restrict their timelines to exclude the missing periods and explicitly handle these anomalies prior to training.